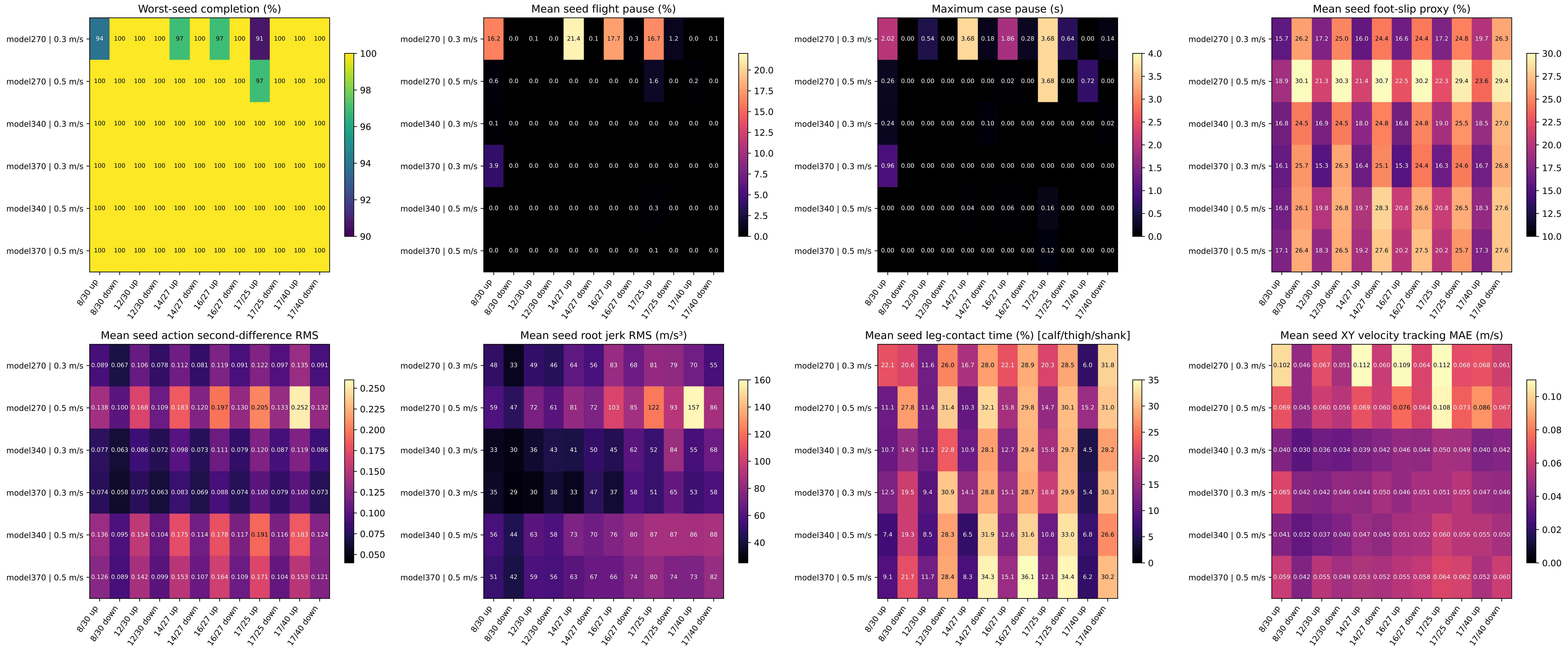
Standard low-speed continuation | matching stair geometry, 3 reset seeds × 32 envs/case | rise/tread cm
New model340 and model370 pass all cases at both 0.3 and 0.5 m/s; motion metrics retain checkpoint and speed tradeoffs.


Worst-seed completion (%)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 94 100 100 100 97 100 97 100 91 100 100 100
model270 | 0.5 m/s: 100 100 100 100 100 100 100 100 97 100 100 100
model340 | 0.3 m/s: 100 100 100 100 100 100 100 100 100 100 100 100
model370 | 0.3 m/s: 100 100 100 100 100 100 100 100 100 100 100 100
model340 | 0.5 m/s: 100 100 100 100 100 100 100 100 100 100 100 100
model370 | 0.5 m/s: 100 100 100 100 100 100 100 100 100 100 100 100
90 92 94 96 98 100
Mean seed flight pause (%)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 16.2 0.0 0.1 0.0 21.4 0.1 17.7 0.3 16.7 1.2 0.0 0.1
model270 | 0.5 m/s: 0.6 0.0 0.0 0.0 0.0 0.0 0.0 0.0 1.6 0.0 0.2 0.0
model340 | 0.3 m/s: 0.1 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
model370 | 0.3 m/s: 3.9 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
model340 | 0.5 m/s: 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.3 0.0 0.0 0.0
model370 | 0.5 m/s: 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.1 0.0 0.0 0.0
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0
Maximum case pause (s)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 2.02 0.00 0.54 0.00 3.68 0.18 1.86 0.28 3.68 0.64 0.00 0.14
model270 | 0.5 m/s: 0.26 0.00 0.00 0.00 0.00 0.00 0.02 0.00 3.68 0.00 0.72 0.00
model340 | 0.3 m/s: 0.24 0.00 0.00 0.00 0.00 0.10 0.00 0.00 0.00 0.00 0.00 0.02
model370 | 0.3 m/s: 0.96 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
model340 | 0.5 m/s: 0.00 0.00 0.00 0.00 0.04 0.00 0.06 0.00 0.16 0.00 0.00 0.00
model370 | 0.5 m/s: 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.12 0.00 0.00 0.00
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Mean seed foot-slip proxy (%)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 15.7 26.2 17.2 25.0 16.0 24.4 16.6 24.4 17.2 24.8 19.7 26.3
model270 | 0.5 m/s: 18.9 30.1 21.3 30.3 21.4 30.7 22.5 30.2 22.3 29.4 23.6 29.4
model340 | 0.3 m/s: 16.8 24.5 16.9 24.5 18.0 24.8 16.8 24.8 19.0 25.5 18.5 27.0
model370 | 0.3 m/s: 16.1 25.7 15.3 26.3 16.4 25.1 15.3 24.4 16.3 24.6 16.7 26.8
model340 | 0.5 m/s: 16.8 26.1 19.8 26.8 19.7 28.3 20.8 26.6 20.8 26.5 18.3 27.6
model370 | 0.5 m/s: 17.1 26.4 18.3 26.5 19.2 27.6 20.2 27.5 20.2 25.7 17.3 27.6
10.0 12.5 15.0 17.5 20.0 22.5 25.0 27.5 30.0
Mean seed action second-difference RMS
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 0.089 0.067 0.106 0.078 0.112 0.081 0.119 0.091 0.122 0.097 0.135 0.091
model270 | 0.5 m/s: 0.138 0.100 0.168 0.109 0.183 0.120 0.197 0.130 0.205 0.133 0.252 0.132
model340 | 0.3 m/s: 0.077 0.063 0.086 0.072 0.098 0.073 0.111 0.079 0.120 0.087 0.119 0.086
model370 | 0.3 m/s: 0.074 0.058 0.075 0.063 0.083 0.069 0.088 0.074 0.100 0.079 0.100 0.073
model340 | 0.5 m/s: 0.136 0.095 0.154 0.104 0.175 0.114 0.178 0.117 0.191 0.116 0.183 0.124
model370 | 0.5 m/s: 0.126 0.089 0.142 0.099 0.153 0.107 0.164 0.109 0.171 0.104 0.153 0.121
0.050 0.075 0.100 0.125 0.150 0.175 0.200 0.225 0.250
Mean seed root jerk RMS (m/s³)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 48 33 49 46 64 56 83 68 81 79 70 55
model270 | 0.5 m/s: 59 47 72 61 81 72 103 85 122 93 157 86
model340 | 0.3 m/s: 33 30 36 43 41 50 45 62 52 84 55 68
model370 | 0.3 m/s: 35 29 30 38 33 47 37 58 51 65 53 58
model340 | 0.5 m/s: 56 44 63 58 73 70 76 80 87 87 86 88
model370 | 0.5 m/s: 51 42 59 56 63 67 66 74 80 74 73 82
40 60 80 100 120 140 160
Mean seed leg-contact time (%) [calf/thigh/shank]
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 22.1 20.6 11.6 26.0 16.7 28.0 22.1 28.9 20.3 28.5 6.0 31.8
model270 | 0.5 m/s: 11.1 27.8 11.4 31.4 10.3 32.1 15.8 29.8 14.7 30.1 15.2 31.0
model340 | 0.3 m/s: 10.7 14.9 11.2 22.8 10.9 28.1 12.7 29.4 15.8 29.7 4.5 28.2
model370 | 0.3 m/s: 12.5 19.5 9.4 30.9 14.1 28.8 15.1 28.7 18.8 29.9 5.4 30.3
model340 | 0.5 m/s: 7.4 19.3 8.5 28.3 6.5 31.9 12.6 31.6 10.8 33.0 6.8 26.6
model370 | 0.5 m/s: 9.1 21.7 11.7 28.4 8.3 34.3 15.1 36.1 12.1 34.4 6.2 30.2
0 5 10 15 20 25 30 35
Mean seed XY velocity tracking MAE (m/s)
8/30 up, 8/30 down, 12/30 up, 12/30 down, 14/27 up, 14/27 down, 16/27 up, 16/27 down, 17/25 up, 17/25 down, 17/40 up, 17/40 down
model270 | 0.3 m/s: 0.102 0.046 0.067 0.051 0.112 0.060 0.109 0.064 0.112 0.068 0.068 0.061
model270 | 0.5 m/s: 0.069 0.045 0.060 0.056 0.069 0.060 0.076 0.064 0.108 0.073 0.086 0.067
model340 | 0.3 m/s: 0.040 0.030 0.036 0.034 0.039 0.042 0.046 0.044 0.050 0.049 0.040 0.042
model370 | 0.3 m/s: 0.065 0.042 0.042 0.046 0.044 0.050 0.046 0.051 0.051 0.055 0.047 0.046
model340 | 0.5 m/s: 0.041 0.032 0.037 0.040 0.047 0.045 0.051 0.052 0.060 0.056 0.055 0.050
model370 | 0.5 m/s: 0.059 0.042 0.055 0.049 0.053 0.052 0.055 0.058 0.064 0.062 0.052 0.060
0.00 0.02 0.04 0.06 0.08 0.10